COLORGRAPH

AMD RADEON HACKATHON 2026 | TRACK 1

Local AI color grading you can inspect, edit, and trust.

ColorGraph turns evidence and creative intent into a typed, reversible Grade Graph, then validates every candidate on real frames before a human chooses what to apply.

ACES 2
Explicit OCIO color pipeline

3-way
Audited creative directions

Fail-closed
Hard QC + human confirmation

Project Profile | 02 August 2026

AMD execution pending - no device claim

PROJECT BACKGROUND

Keep the camera original. Make the grade explainable.

Creators need the speed of AI without surrendering color science, source integrity, or editorial control.

The gap

- Manual grading is precise, but slow to explore across shots, masks, and keyframes.
- Black-box generative workflows can be fast, but may redraw detail, hide transforms, or make revisions hard to reproduce.
- Professional log, high-bit-depth, VFR, and scene-scoped work need explicit technical boundaries.
- AI suggestions should remain proposals until measured evidence and a human decision agree.

COLORGRAPH PRINCIPLE

AI proposes.
The graph records.
QC verifies.
The creator decides.

No candidate silently mutates the current grade.

Target users and scenarios

Independent creators

Fast first-pass balance, look exploration, natural-language revisions, and a portable delivery package.

Small post teams

Deterministic candidate review, scoped changes, consistent scenes, quality evidence, and exact undo.

Education / evaluation

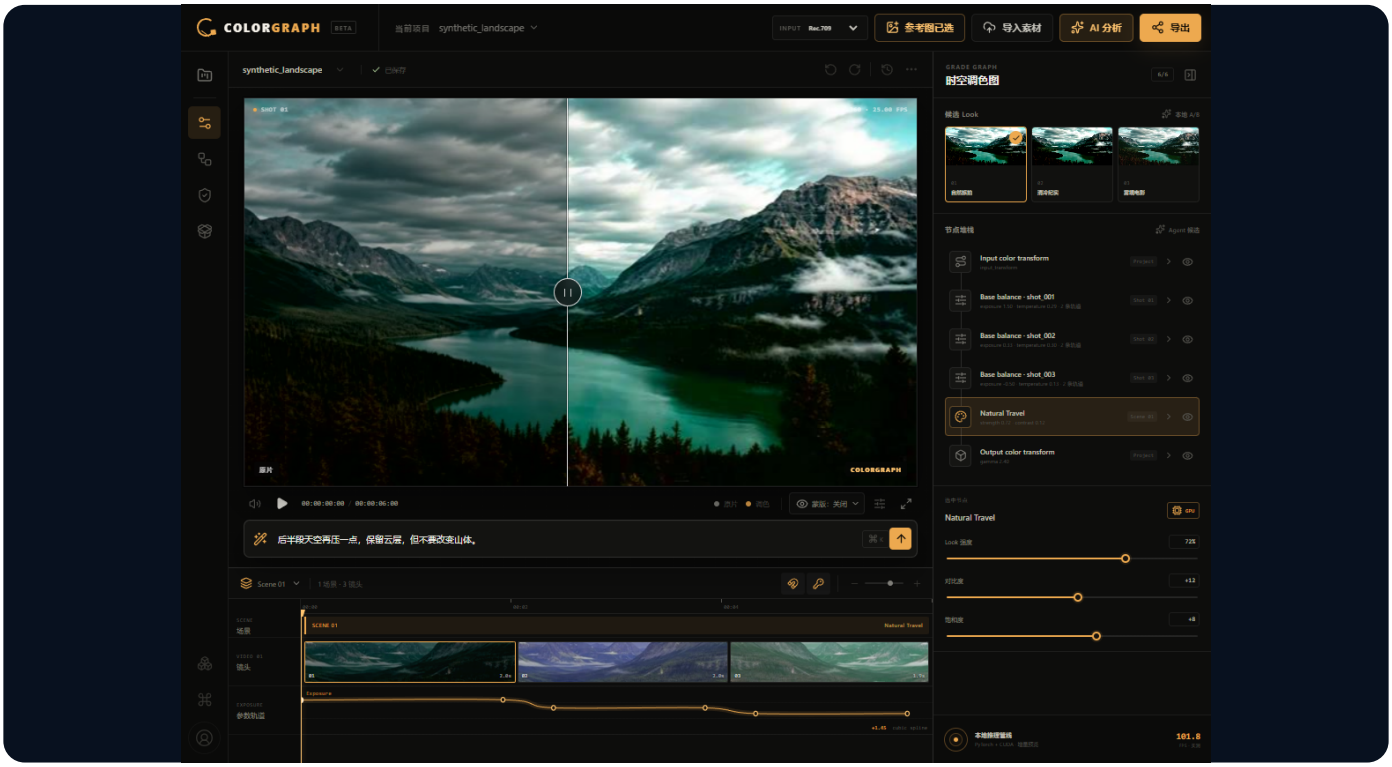
An inspectable Grade Graph, fixed color transforms, frozen assets, and metrics that separate engineering from release claims.

Outcome A local workstation that accelerates creative exploration while leaving a reversible, reviewable technical record.

PRODUCT EXPERIENCE

One workstation from evidence to delivery

The interface keeps source, graded preview, timeline, masks, keyframes, Agent candidates, graph nodes, and export evidence in the same context.



Current local workstation UI (Chinese localization); this Project Profile is the English submission artifact.

Creator flow

01 — Import
Local path or upload

02 — Analyze
Shots + color evidence

03 — Explore
Looks + 3 Agent directions

04 — Refine
Masks + keyframes + language

05 — Deliver
Render + QA + package

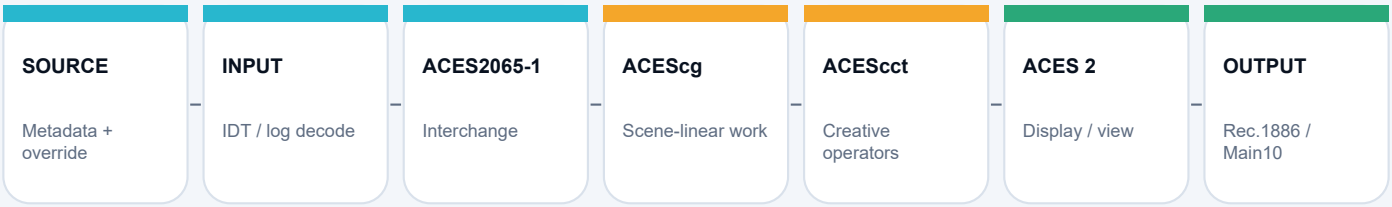
REVERSIBLE BY DESIGN

Every applied command has a precise scope and undo record. Candidate search, playback previews, and final rendering are content-addressed so stale work is invalidated rather than reused silently.

CORE ARCHITECTURE

Color science stays explicit end to end

ColorGraph does not hide color management inside a preset. Transforms, creative operations, scopes, and output intent are versioned graph data.



Typed Grade Graph

- Strict Pydantic graph contracts with renderer and schema versions.
- Project, scene, shot, sub-shot, time, global, mask, and composite scopes.
- Exposure, CCT/Duv balance, CDL, contrast, saturation, highlights, curves, LUTs, and PCHIP keyframes.
- Scene-isolated looks and content-addressed preview invalidation.
- Exact graph diff, candidate contract, undo graph, and export manifest.

High-bit-depth render path

- FFprobe preserves range, primaries, transfer, matrix, depth, chroma, HDR, and evidence confidence.
- rgb48le handoff and FP32 grading for the formal render.
- CFR streaming or VFR PTS reconstruction with frame/audio round-trip gates.
- HEVC Main10 hardware selection where verified; software fallback remains explicit.
- Post-render QA can apply one bounded, targeted repair and re-verify.

Local perception and temporal structure

Shots

Current detector + explicit shadow alternatives

Scene

CLIP-based evidence and optional local VLM path

Regions

Semantic + depth anchors with temporal propagation

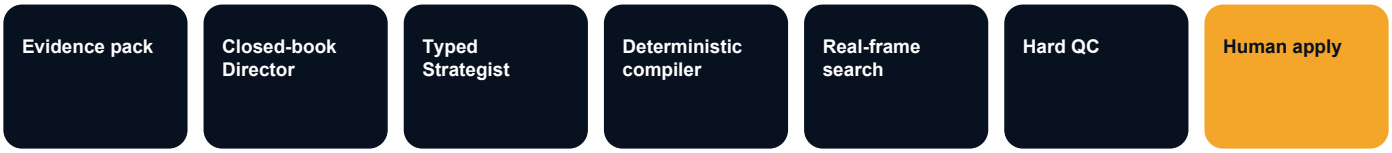
Quality

Highlights, shadows, gamut, structure, flicker, masks

AI AND ALGORITHM DESIGN

Creative intelligence inside a deterministic boundary

The Agent reasons over signed evidence and typed capabilities. It cannot invent renderer nodes, bypass QC, or apply a result without confirmation.



Frozen direction grammar

FAITHFUL

0.00 nominal

Contrast preserve

Keep the source relationship primary.

AMPLIFIED

+0.30 to +0.45

Contrast increase

Strengthen hierarchy while retaining detail.

COUNTERPOINT

-0.55 to -0.40

Saturation decrease

Create quiet tension without erasing structure.

Why the boundary matters

- Evidence IDs and SHA-256 inputs are immutable.
- Capability catalog is generated from the current Grade Graph.
- Direction signature and effect floor reject near-neutral convergence.
- Compiler preserves input/output transforms and emits exact undo.
- Candidates render the same frozen sample frames.
- Asset, schema, finite-value, and output-gamut gates are non-waivable.
- Provider and Critic stages are optional, budgeted, and audited.
- Automatic selection stays off without independent expert calibration.

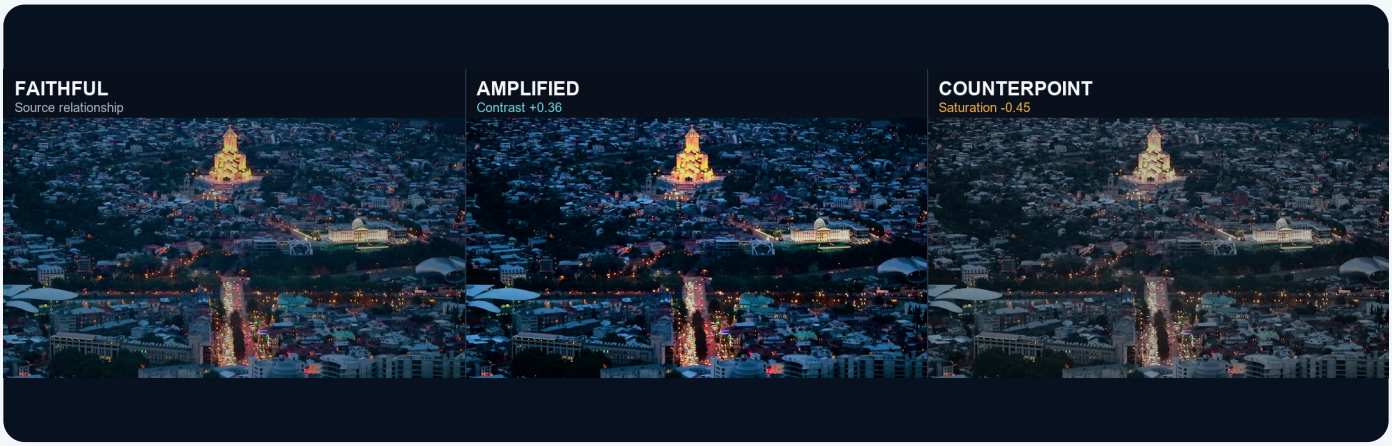
REAL-FRAME ENGINEERING CHECK

Tbilisi offline run: 3 strategies compiled, 9 samples rendered, 3/3 candidates hard-pass, 0 illegal-output samples, current graph unchanged.

PUBLIC DEMO STRATEGY

Use footage that makes the grade visible

A public CC BY-SA lane supports reproduction; the premium DJI lane proves log/high-bit-depth handling locally without redistributing the source.



Modified stills from Tbilisi life at night - Mostafameraji - CC BY-SA 4.0 | Values shown are typed strategy parameters.

ROLE	SOURCE	VISUAL VALUE	PUBLIC RULE
Creative hero	Tbilisi night	Warm/cool city separation	CC BY-SA; attribute + share alike
Workflow	Snowdonia drone	Cuts, terrain, water, road	CC BY-SA; hash-verified download
Log proof	DJI M3 Pro Cine	4K ProRes 10-bit D-Log	Local only; finished display needs terms

Rights question

Does the agreement allow modification and public/commercial display of the finished work? Does it separately allow redistribution of the raw source or an extract?

Size question

Even with redistribution rights, the full DJI source is 25.68 GB and a five-second ProRes extract is about 1.05 GB. Neither belongs in normal Git history.

VIDEO DELIVERABLE PAUSED

This profile freezes footage and visual direction only. No final 3-5 minute demo video has been created.

AMD ADAPTATION

Portable architecture, fail-closed Radeon proof

The compute boundary is designed for PyTorch ROCm. The project does not convert CUDA preflight results into AMD claims; a supported Radeon host must produce the final evidence.

STATUS **Engineering boundary implemented; real AMD Radeon / ROCm execution is pending and intentionally paused.**

Already in the architecture

- ComputeBackend exposes device, inference context, synchronization, cache release, and structured identity.
- TorchComputeBackend distinguishes HIP from CUDA through torch.version.hip while using standard PyTorch tensor APIs.
- Core grading and model paths avoid custom CUDA, Triton, FlashAttention, xFormers, and .cu extensions.
- Windows ROCm encoding selects H.264/HEVC AMF only after preflight; software fallback remains explicit.
- Stage evidence rejects non-ROCm runtime, non-AMD vendor, empty HIP identity, or unverified tensor placement.

Required on the target Radeon

1. Verify supported GPU, driver, ROCm/HIP, PyTorch wheel, FFmpeg, and AMF identity.
2. Run doctor and three-repeat CPU FP32 vs ROCm FP32 precision gate.
3. Run local CLIP, semantic, depth, and configured VLM stages with GPU placement evidence.
4. Fresh-render the public frozen clip, including mask and no-mask rows.
5. Verify HEVC Main10, timestamps, masks, quality report, hashes, warm FPS, latency, and peak VRAM.
6. Repeat the complete run and retain the signed, content-bound evidence package.

Honest execution boundary

RADEON / ROCm GPU CLIP | semantic | depth | VLM | Grade Graph | LUT | mask blend | ACES 2 output

MEDIA / CPU BOUNDARY Demux | high-bit-depth decode | audio mux | file hash | post-render measurement

Release rule: only evidence whose runtime=rocm, vendor=AMD, HIP identity, tensor placement, encoder preflight, media hashes, and quality gates all agree may be described as AMD-verified.

COLORGRAPH

DELIVERY AND READINESS

Reproducible now. Submission-complete later.

The non-AMD package is prepared without hiding the remaining competition gates.

Prepared

- English source README and architecture
- Public media downloader + immutable hashes
- Third-party attribution and DJI rights boundary
- Visible three-direction strategy contract
- Real-frame hard-QC evidence
- Non-AMD CI and regression commands
- English Project Profile PDF

Still required

- Project source-code license decision
- Actual DJI authorization review (if used)
- Supported Radeon / ROCm full execution
- Measured AMD FPS / latency / VRAM evidence
- Final 3-5 minute demo video
- PPT or Poster
- Public repository + official contest PR

Public reproduction

```
.\scripts\fetch-demo-assets.ps1 -Asset all
uv run --project backend colorgraph analyze .\demo-assets\tbilisi-night-canon-eos-m3.ogv --skip-scene-model
npm run build
uv run --project backend pytest
```

SOURCE AND DOCUMENTATION

github.com/panbu2007/color | README.md | docs/SUBMISSION_CHECKLIST.md | docs/DEMO_ASSETS.md

ColorGraph makes the creative path faster without making the evidence disappear.